WINDOWS SERVER 2025
ACTIVE DIRECTORY HOME LAB

DAY 2 — Active Directory, DNS, Identity Management & NTFS Permissions

Lab date: October 22, 2026 | Platform: Oracle VirtualBox | Server: DC01

Executive summary

Configured a dedicated internal lab network, installed and promoted Active Directory Domain Services on Windows Server 2025, created domain users and role-based security groups, applied NTFS permissions to an HR documents folder, reviewed effective access, and saved a Day 2 VirtualBox snapshot. All work was performed in an isolated learning environment.

Environment and configuration

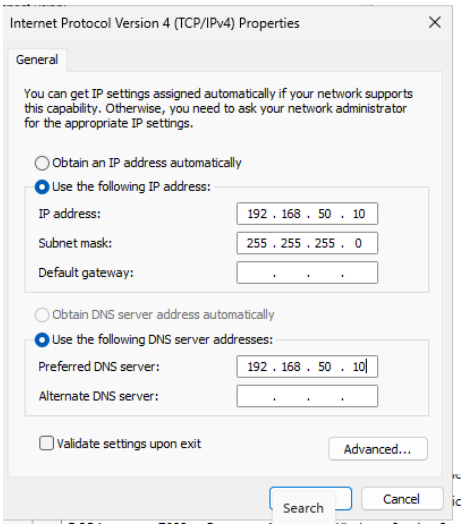
Component	Observed configuration
Operating system	Windows Server 2025 Standard Evaluation
Server / domain	DC01 / adlab.test (NetBIOS: ADLAB)
VirtualBox internal network	ADLAB, Adapter 2
Internal IPv4 / mask	192.168.50.10 / 255.255.255.0
Internal DNS	192.168.50.10
Other adapter	NAT/DHCP for internet access
Installed roles	Active Directory Domain Services; DNS

Key outcomes

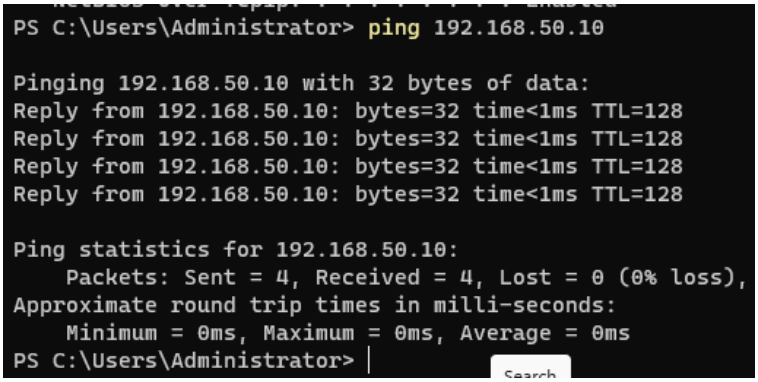
1. Verified static addressing and network connectivity.
2. Created the adlab.test forest and promoted DC01 to a domain controller.
3. Created the Lab-Users OU, two test users, and two departmental security groups.
4. Restricted C:\HR-Documents using explicit NTFS permissions.
5. Examined Effective Access and captured a Day 2 recovery snapshot.

1 | Internal network and connectivity

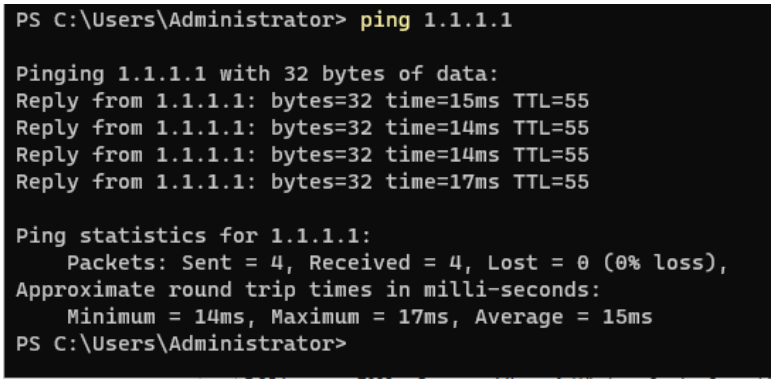
Added a VirtualBox internal network adapter named ADLAB. Configured Ethernet 2 with static address 192.168.50.10/24, no internal default gateway, and DNS 192.168.50.10. A separate DHCP-connected adapter continued to provide internet access. The screenshots show successful ping replies to both 192.168.50.10 and 1.1.1.1 (four replies, zero loss in each test).



Evidence 1 — Static IPv4 and DNS configuration on the internal adapter.



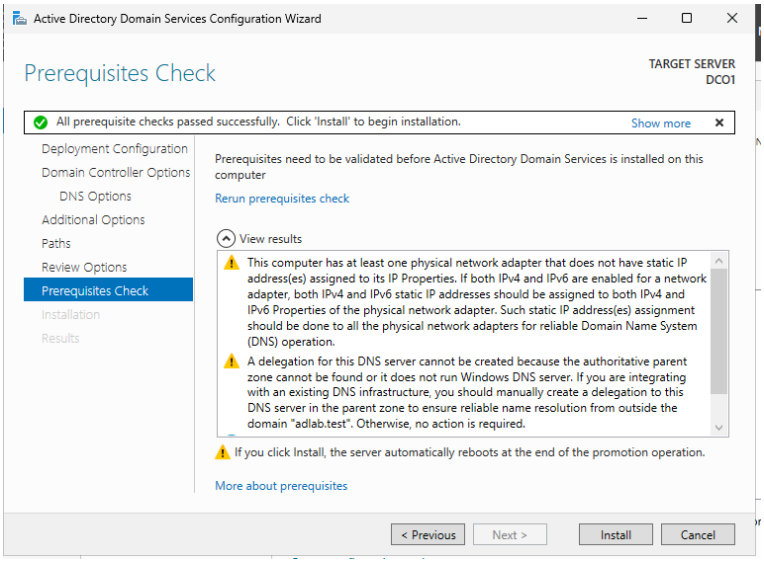
Evidence 2 — Successful local IPv4 ping test to 192.168.50.10.



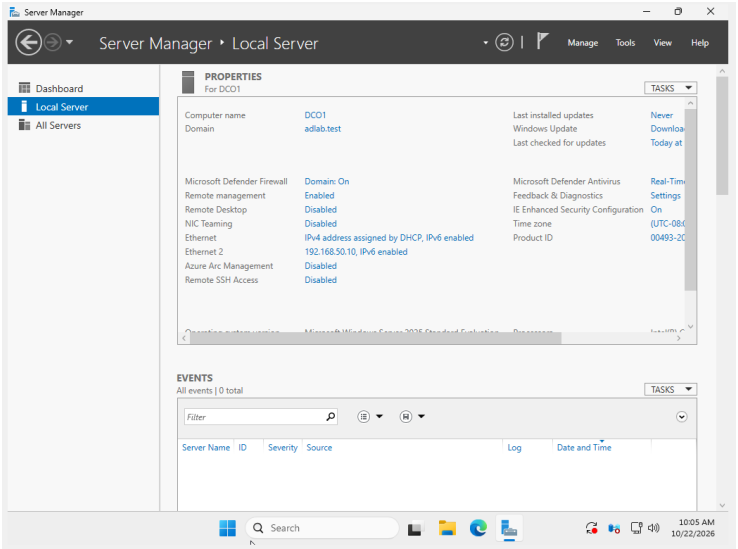
Evidence 3 — Successful internet connectivity test to 1.1.1.1.

2 | Active Directory deployment

Installed Active Directory Domain Services and its management tools, then promoted DC01 as the first domain controller in a new forest named adlab.test. The deployment review showed DNS and Global Catalog enabled and Windows Server 2025 domain/forest functional levels. Prerequisite checks passed; the wizard also displayed warnings about a network adapter without a static IP and the absence of a DNS delegation. The DNS delegation warning is expected for a standalone new lab forest. The server restarted after promotion.



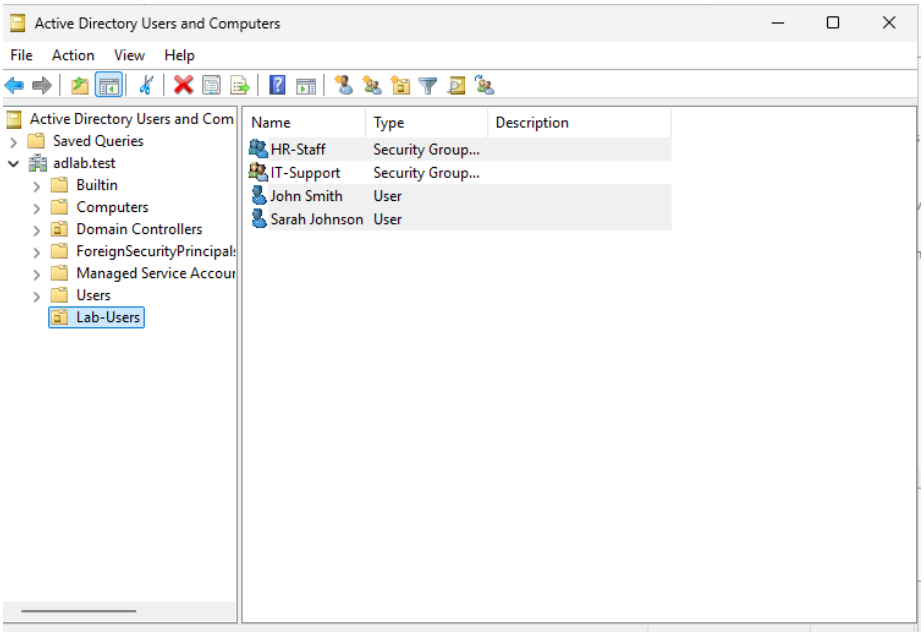
Evidence 4 — Prerequisite checks passed, with recorded configuration warnings.



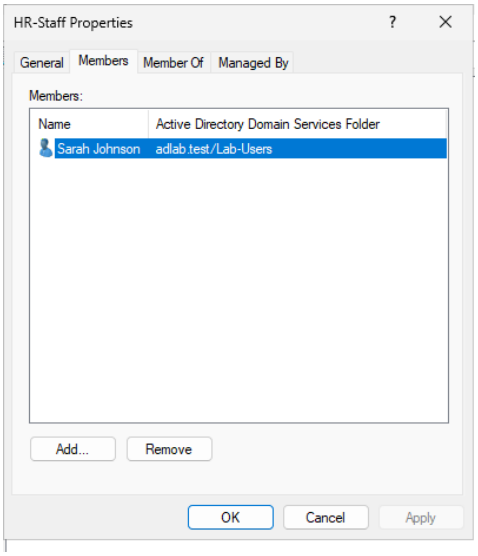
Evidence 5 — Server Manager confirms domain membership and internal address.

3 | Organizational units, users, and security groups

Created the Lab-Users organizational unit under adlab.test. Added test users John Smith (jsmith) and Sarah Johnson, plus IT-Support and HR-Staff security groups. Membership screenshots show John Smith in IT-Support and Sarah Johnson in HR-Staff. John Smith was configured to change his password at next logon.



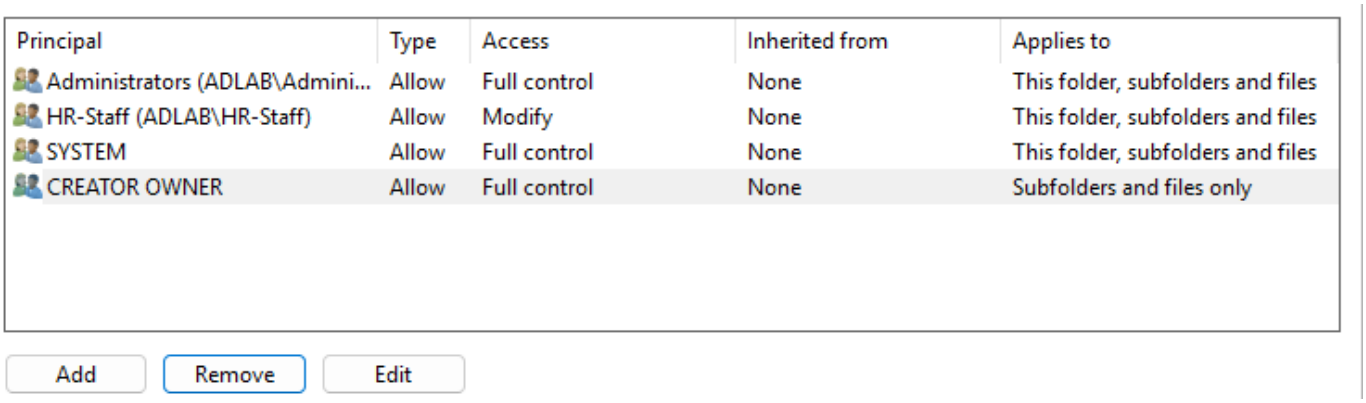
Evidence 6 — Lab-Users OU with both test users and departmental groups.



Evidence 7 — HR-Staff group membership includes Sarah Johnson.

4 | NTFS access control for HR documents

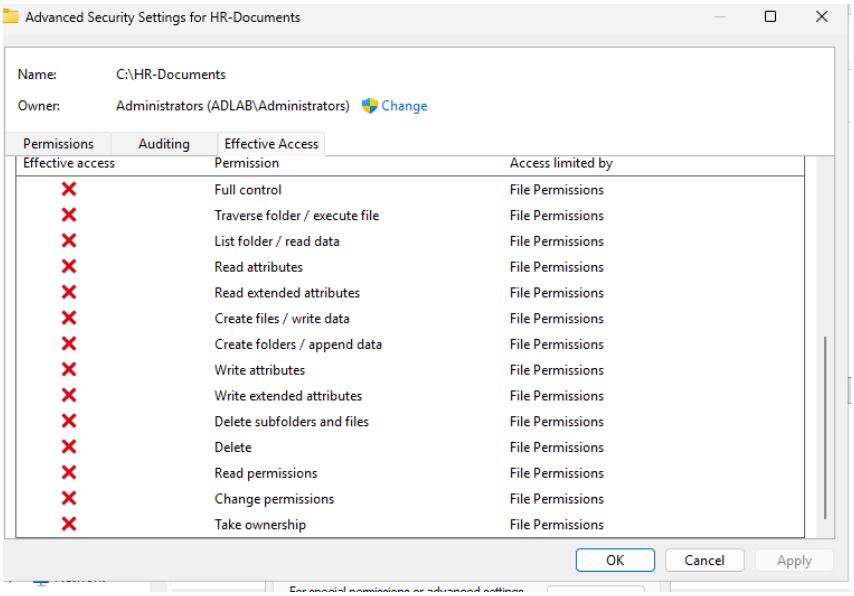
Created C:\HR-Documents and applied explicit permissions. The final permissions view shows Administrators and SYSTEM with Full Control, HR-Staff with Modify access, and CREATOR OWNER with inherited-object Full Control. Inheritance was disabled and broad Users entries were removed. These settings demonstrate group-based access control and least-privilege design in a training environment.



Evidence 8 — Final explicit NTFS access control entries for C:\HR-Documents.

5 | Effective Access review

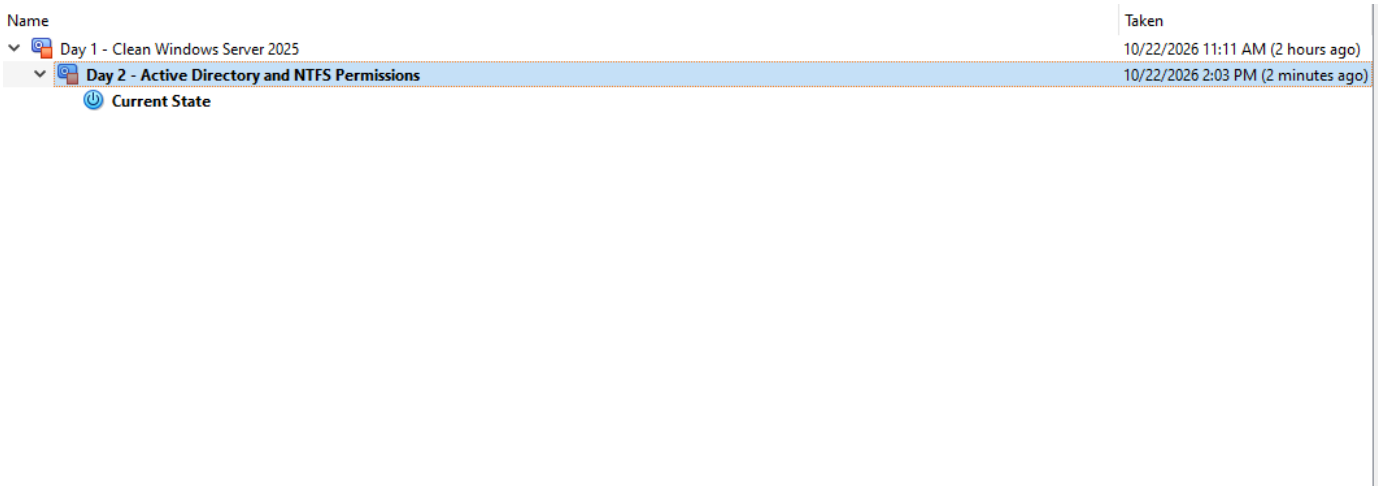
Used Advanced Security Settings > Effective Access to inspect the permissions available to selected identities. Captured screenshots show both permitted and denied operations; the denied-access screenshot is retained below. These checks are not a substitute for a full sign-in and file-opening test as the target user.



Evidence 9 — Effective Access displays denied operations for the tested identity.

6 | Snapshot and recovery checkpoint

Saved the VirtualBox snapshot named **Day 2 - Active Directory and NTFS Permissions**, following the earlier Day 1 clean Windows Server snapshot. This provides a rollback point before proceeding to Group Policy configuration.



Evidence 10 — VirtualBox snapshot tree confirms the Day 1 and Day 2 checkpoints.

Technical skills demonstrated

Windows Server administration; VirtualBox networking; TCP/IP and DNS; Active Directory Domain Services; domain controller promotion; organizational unit management; user provisioning; security group membership; NTFS discretionary access control lists (DACLs); inheritance; effective permissions; troubleshooting; and recovery snapshots.

Validation and limitations

Verified: static internal IPv4 configuration, ping replies, successful AD DS installation and domain promotion, OU/users/groups visible in Active Directory Users and Computers, configured NTFS access entries, Effective Access checks, and snapshot creation.
Further validation recommended: sign in as test users and confirm actual allowed/denied access to C:\HR-Documents; check DNS name resolution for adlab.test; review the static-IP prerequisite warning and ensure the domain controller uses a stable DNS configuration.

Day 3 planned work

Configure Group Policy Objects, password and account lockout settings, apply security policies to appropriate OUs, and verify policy application with gpresult or equivalent tools.

Documentation is based on screenshots captured during the Day 2 lab. This is a personal training environment, not a production deployment.